

LEStrADE: Investigating Account Compromise using Data Exports for Survivors of Technology-Facilitated Abuse

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Abstract

Compromise of a survivor’s online accounts is a pervasive vector of technology-facilitated abuse (TFA). The built-in account security interfaces provided by online platforms often expose limited and incomplete information and use ambiguous device identifiers, making it difficult for survivors to determine whether, when, and by whom their accounts may have been compromised. To address this gap, we present **LEStrADE** (Local Engine for Structured Analysis of Data Exports), a browser-based digital forensics tool that locally parses and visualizes survivors’ account data exports so that they can identify and annotate potentially suspicious account activity. LESTrADE includes an entity resolution heuristic that identifies events originating from the same browser or app instance, achieving 84–98% precision on real browser data. We evaluate LESTrADE through an iterative user study with 11 technology clinic consultants who directly support TFA survivors. Their feedback demonstrates how personal data exports can provide otherwise inaccessible evidence to support investigations of suspected account compromise.

1 Introduction

Online accounts make up much of the infrastructure through which people conduct their digital lives. Simultaneously, online accounts can open channels of harassment, control, and surveillance for at-risk users who, due to contextual risk factors, experience elevated threats to their digital safety and security [96, 107]. For survivors of intimate partner violence (IPV), account compromise is a common tactic of technology-facilitated abuse (TFA), as abusers (intimate partners) can leverage shared passwords and physical proximity to devices to gain authenticated access to their online accounts [42].

When a survivor suspects that one of their online accounts has been compromised, their typical way to triage the situation is to visit *account security interfaces* (ASIs) [16, 33] such as Google’s *Your devices* page [47] or the *Where you’re logged in* within Meta’s Accounts Center [71]. Although ASIs

are valuable sources of information, they have several known flaws. Crucially, they underreport historical logins and active sessions. For example, Google shows security events only from the last 28 days, and Apple often fails to display some browser logins to apple.com [16, 33]. Even when relatively complete, ASIs can be difficult to interpret with enough certainty to judge if activity is malicious or benign. Daffalla et al. [33] observed that both TFA survivors and technologists struggle to make sense of ambiguous device identifiers (i.e., model or operating system) or multiple sessions on a single device. Judgments often depend on contextual information outside the ASI itself, such as whether a login predates when the abuser last had physical access to the survivor’s phone [54].

However, Nonnenkamp et al. [73] suggest that personal *data exports* can partially address the issues of completeness and underreporting in ASIs. Data exports are copies of users’ personal account data that companies must provide on request due to data protection regulations around the world [23, 39, 55]. Compared to ASIs, data exports tend to provide longer authentication logs and more detailed device identifiers (e.g., IP addresses, session cookies) [73].

Usually, without a subpoena, users have no comparable way to obtain this volume and depth of information to investigate an account compromise. However, data exports — usually ZIP archives with dozens or hundreds of files — are difficult to navigate and interpret [18, 102]. No prior work has realistically operationalized them to investigate account compromise, particularly when responding to TFA. Existing data export visualization tools [9, 17, 18, 82, 87] were built to support transparency-related goals (i.e., exposing ubiquitous data collection), *not* for helping at-risk users, particularly survivors of IPV or TFA, investigate account compromise.

In this paper, we present **LEStrADE**¹ (Local Engine for Structured Analysis of Data Exports), a browser-based tool that lets users analyze their data exports and annotate potential account compromise.² Currently, LESTrADE is designed

¹Named after the Sherlock Holmes character, Inspector Lestrade.

²Repository available at <https://anonymous.4open.science/r/data-export-gui-10F4/>.

primarily for technologists supporting TFA survivors through the *technology clinic* model [43, 54], although it may be useful for other at-risk users beyond this context. LEStADE currently supports six major platforms: Apple, Google, Facebook, Instagram, Snapchat, and Discord; support for others can be added.

LEStADE is a static web application that processes data exports entirely within the local browser file system so that it can be used without installing new software. It creates a normalized representation of security information in data exports so that users can view, filter, and search records from multiple platforms in a unified timeline (“Events View”). To help users understand which devices have signed in or what actions each device performed (one issue of ASIs [33]), we developed an entity resolution heuristic that groups records that likely originate from the same physical device or client instance (i.e., browser, mobile app). LEStADE presents these group abstractions separately (“Devices View”) and lets users annotate data with their interpretation or external context.

We measure the accuracy of the entity resolution heuristic on a subset of the AmIUnique dataset of browser fingerprints [61, 101], as well as the speed and memory use of the data import pipeline. Finally, we conducted a mixed-methods user study with 11 technology clinic volunteers or staff (*consultants*) to understand how they use LEStADE to investigate a hypothetical account compromise scenario with real data exports, and we describe how we iteratively updated the tool to respond to feedback.

Contributions:

- We design and evaluate LEStADE, a tool for investigating account compromise in TFA contexts using data exports. Our entity resolution heuristic achieves 84–98% precision when grouping real browser events by their originating instance using a limited set of device identifiers.
- We present empirical findings from a user study with 11 technology clinic consultants about the strategies they use to investigate account compromise and navigate uncertainty. LEStADE yields favorable usability scores, and we report on challenges, feature suggestions, and deployment considerations.

2 Background & Related Work

We provide background on technology-facilitated abuse and interventions for it, as well as data exports and right of access regulations. Finally, we review related tools and demonstrate why LEStADE offers a novel solution.

2.1 Technology-Facilitated Abuse & Interventions

In contrast to state-sponsored cyberattacks or online scams, *technology-facilitated abuse* (TFA) describes digitally medi-

ated harassment, surveillance, and coercion where the victim-survivor and adversary have an interpersonal relationship [59]. TFA is most frequently discussed in the context of intimate partner violence (IPV) [31, 37, 41, 89], and it is understood to be interwoven with physical, economic, and psychological harms experienced offline [15, 59]. In a study of college students, Gupta et al. [51] found that 70% of respondents had experienced TFA, and among those, 29% experienced it from an intimate partner. Reed et al. [81] found that 77% of 377 IPV survivors had their phones searched without permission and 60% were forced to share passwords. A growing body of security research has characterized how everyday technology can be misused in this threat model [7, 27, 34, 58, 91, 111]. *Account compromise* (i.e., of email, social media, and banking accounts) is a common vector of TFA [41, 81, 96] and the focus of this paper.

Clinical Computer Security Survivors, and those supporting them, respond to TFA in a variety of ways, formal and informal [5, 52, 90, 99]. One prominent formal intervention is the *clinical computer security* model, first introduced by Havron et al. [54] via the Clinic to End Tech Abuse (CETA) in New York City [28]. This framework proposes *technology clinics*, which partner with IPV shelters and advocacy organizations to provide individualized cybersecurity *consultations* to survivors. During a consultation, a trained *consultant* helps a survivor identify compromise of devices and accounts, and then create a digital safety plan alongside the survivor’s *advocate* (i.e., case manager, legal support) [43, 54, 98]. Since CETA’s inception, new technology clinics continue to be developed in the U.S. [1, 32, 94] and Canada [2]. A related line of work explores how technology clinics can improve their services from survivors’ and practitioners’ perspectives [68, 77, 78, 92].

Detecting Account Compromise Because account “hacking” is one of the most commonly reported vectors of abuse [42], much of consultants’ work involves helping survivors navigate and interpret *account security interfaces* [16, 33, 41, 54] such as Google’s *Your Devices* page [47] or Apple’s *Safety Check* feature [10] to manage active sessions or devices and recovery settings. However, these interfaces are difficult for users to find [26, 44], and prior work has found that ASIs underreport login and session information, particularly if a login seems “known” or falls outside of a certain time window [16, 34]. Studying data exports, Nonnenkamp et al. [73] found that platforms *do* tend to record longer and fuller authentication histories than they make available through ASIs. Still, even when present, login details can be hard to interpret. Studying support sessions between technologists and IPV survivors, Daffalla et al. [33] observed that ambiguous device identifiers (i.e., model or operating system) in ASIs can make it difficult to determine if a device listed as “trusted” is benign or not. For this, Ortega Pérez et al. [74] proposed a protocol, “client-side encrypted access logging (CSAL),” to solve this problem by adding an encrypted device

identifier, generated by the OS, to the user’s account activity log that cannot be used to fingerprint the origin device. Though less stable, data exports tend to contain more granular device identifiers than ASIs as well [73].

2.2 Data Exports

Data exports (also known as “Data Subject Access Requests” [18, 64] or “data downloads” [102]) are copies of users’ own personal data that they can request from companies under data protection laws around the world. Data exports are typically ZIP archives containing text (e.g. CSV, JSON, HTML) and media files in a structured folder system.

Legal Basis Article 15 of the European Union’s GDPR [39] provides users (“data subjects”) with *right of access*, or the right to request their personal data from organizations or companies that collect it. The *right of portability* demands that data should be provided in machine-readable and transferable formats [110]. Subsequent data protection laws with similar protections include Brazil’s LGPD [72], India’s DPDP [55], and California’s (U.S.) CCPA/CPRA [22, 23]. Often, online platforms (especially those with multinational footprints) let *all* users exercise their data access rights regardless of regulatory coverage [56].

Obtaining Data Exports Many large platforms provide specific web forms for users to make data request, for example Google’s *Takeout* and Meta’s *Export your information* pages [48, 70]. Processing and delivery of requested data can take several days depending on the platform, and users are usually notified via email when their request is complete [64, 76, 102]. However, both the process of requesting data and the resulting files are often unusable. Prior work has found “dark patterns” and slow processing times in data request flows [66, 76], and outputs occasionally violate machine-readability, or *portability*, requirements set by many jurisdictions [76, 110]. Users also struggle to understand the data they have received [18, 19, 102]. Through co-design, Veys et al. [102] synthesized design requirements for more usable data exports (e.g., chronological organization, aggregation) that help users exercise both the letter and the spirit of their *right of access*.

2.3 Related Tools

Tools for TFA Survivors Tech clinic consultants and investigators routinely use tools specifically designed to assist survivors. For example, the IPV Spyware Discovery (ISDi) toolkit [54], WARNE [67], and SpyGuard [4] scan devices for common stalkerware apps [6, 29]. Chen et al. [30] have called on researchers and developers to employ principles of *trauma-informed computing* (e.g., safety, enablement) when designing for this population. One such tool is Sherlock, developed by Stephenson et al. [93] and designed with input

from legal experts, to help consultants collect digital evidence of TFA on behalf of survivors in a trauma-informed manner. Sherlock take authenticated screenshots of logged-in malicious devices [93], but these remain subject to the same UI limitations identified by Daffalla et al. [33] and others [16, 73, 74]. LEStRADE is the first software tool to extend consultants’ ability to identify account (rather than device) compromise beyond platforms’ native account security interfaces.

Data Export Visualizations Prior work has developed user interfaces for data exports, but with the primary goal of supporting privacy- and transparency-related projects [9, 45, 46, 87]. Those most applicable to our study are “TransparencyVis” by Schufirin et al. [87], a privacy awareness tool for data exports, and a research tool by Borem et al. [18] that lets users annotate their files with reactions. Other tools have been built to solicit data export donations from research participants [35, 79, 80]. We developed a new tool for several reasons. The tools most relevant to our use case lacked public source code linked to papers. Transparency tools typically show users *all* their data, but much of this data consists of noisy system logs that can obscure security threats [73]. Moreover, TFA survivors may be re-traumatized by abusive content in their data [30]. We discuss unique design considerations for our context in § 3.3.

Digital Forensics Tools Investigators with law enforcement or security teams use forensic tools to extract digital artifacts like disk images, cloud backups, and subpoenas or warrants from cloud providers [53, 57, 88]. Open source tools include Plaso and Timesketch for timestamped logs [49, 50, 75], Amnesty International’s Mobile Verification Toolkit for detecting commercial spyware [8], and others [14, 20, 24, 25, 36, 104]. There is extremely limited open-source software for analysis of user *account* data. As of this writing, we have only identified two tools in this category. The first, RLEAPP [20, 21], provides basic parsing support for some warrant formats and Google Takeout, but has no semantic model for what any given data entry *means*. The second, the data export parsing scripts by Nonnenkamp et al. [73], classify security-related records into a basic taxonomy, but are brittle to changing schemas and do not standardize fields like timestamps or IP addresses. LEStRADE attempts to address these limitations.

3 Designing LEStRADE

Data exports contain valuable information for identifying account compromise, especially in TFA contexts [73]. However, their size, heterogeneous structure, and sensitive nature make them difficult to use during an investigation [18, 102]. To operationalize data exports for account compromise investigation of TFA survivors, we developed LEStRADE and protocol to integrate it with tech clinics. In this section, we discuss our design process and the considerations that shaped the tool.

3.1 Threat Model & Technical Scope

We consider an adversary (*abuser*) has obtained authenticated access to a victim’s (*survivor’s*) online account with the goal of surveillance or coercive control. The abuser may know a password, use physical device access to learn authentication codes, or create persistent channels of access using recovery methods or passkeys. Consistent with prior work on TFA, we assume that most abusers are *UI-bound*, meaning that they misuse features in the user interfaces and do not use sophisticated exploits [41].

The survivor’s main goal is to identify if unauthorized access has or is happening and, if so, understand its timing or consequences. They may have already visited their account security interface (ASI) and been unsatisfied with the information present (see § 2.1; [16, 34, 73]).

Any tool that ingests data exports is inherently constrained by the data that platforms themselves log and make available. For example, if an abuser spoofed their user agent while authenticating (unlikely as most are UI-bound), this might produce misleading logs in the data export. This is not unique to LESTrADE; this would propagate to the platform’s ASI as well [33]. Client-side encrypted access logging (CSAL) [74] offers a more robust approach to device identification, but has not been adopted by platforms. Data exports may not be able to definitively determine which *person* was responsible for a logged action if multiple people were signed in or using a single device.

3.2 Intended Operational Protocol

Positionality & Design Lens. The challenge of detecting unauthorized account access in TFA contexts is well established [34, 43, 54]. The authors have more than a decade of combined experience working directly with 100+ survivors of intimate partner violence (IPV) who experience TFA, as well as with their case managers and legal advocates. We draw on that experience and adopt the trauma-informed computing lens introduced by Chen et al. [30]. In the context of our tool, *enablement* and *collaboration* require that survivors retain meaningful agency when interpreting their own data. Whether a login is “authorized” or not is often contingent on external context like the issuance of a restraining order or which days of the week the abuser has custody of a shared child and their electronic devices. *Safety* requires us to consider what content might be re-traumatizing and what responses (e.g., changing a password) might cause retaliation.

Collaborative Use in a Technology Clinic. We designed LESTrADE primarily to be used **collaboratively by a tech clinic consultant and a survivor** during a consultation. Consultants already use software tools to detect spyware and document evidence [54, 93], and their involvement can help survivors request and interpret exports safely. A tech clinic consultation with a survivor usually includes a basic account

security check using ASIs [54]. If the ASI does not adequately explain or document suspected compromise [33], or if more detailed investigation is desired, the consultant and survivor may decide to look at the account’s data export. In our experience, a few clients have already downloaded their data exports, although it is more likely that the consultant and client will need to request it together. Because platforms may take hours or days to prepare an export [66], use of LESTrADE will often occur during a follow-up consultation. In these cases, the consultant would provide instructions for safely storing the files. At the next meeting, the survivor and consultant can examine and annotate security records with LESTrADE, then incorporate findings into safety planning.

This is our intended protocol, but not one we have yet fully validated in a clinical setting. We evaluate consultants’ use of the tool during a simulated investigation and our findings include additional deployment considerations (§ 6.5).

3.3 Design Decisions

The threat model, operational workflow, trauma-informed design lens, and known limitations of data exports led to four initial design decisions:

- (a)  **Process data exports locally.** Because data exports often contain sensitive personal information, they should remain on a device the survivor is comfortable using. This also lets survivors or consultants delete the working data after using the tool. This follows the privacy-preserving approach of prior tools for TFA survivors [4, 54, 93].
- (b)  **Minimize exposure to non-security data.** We limit LESTrADE’s scope to information identified as “security-relevant” by prior work, including authentication activity and changes to critical account settings [73]. Although other data (e.g., messages, search history) can help build a timeline, it can expose survivors to private or uncomfortable content [18], especially if using the tool with another person, as is our intent.
- (c)  **Standardize records from different platforms with provenance.** Platforms use different filenames and terminology for comparable actions. Mapping records to standard fields and semantic categories lets users search, filter, and compare activity across platforms, following design recommendations for usable data exports [102]. The tool should retain references to the original record, and the interface must distinguish between information reported by a platform from relationships inferred by LESTrADE.
- (d)  **Support contextual interpretation.** Automatic classification of records as benign or malicious, like an intrusion detection system, could cause false alarm or a false sense of safety because most methods rely on data that lose signal in TFA contexts (e.g., IP addresses may be moot if the survivor and abuser share a home network) [109]. Identifying account compromise often depends on exter-

nal information known only to the survivor (see § 3.2). Our tool should make analysis more usable while leaving the final interpretation to the survivor.

These are design commitments grounded in the deployment context rather than hypotheses tested independently in our study. Instead, our evaluation examines how technology clinic consultants investigate account activity with LESTrADE, where uncertainty remains, which interface features create friction, and what must be considered before deployment (§ 6).

System Overview. LESTrADE has three components that realize the design decisions. First, the *ingestion and normalization* component processes files entirely within the browser, which isolates “security-relevant” records, maps records to a common schema, and preserves source information (§ 4.1). Second, the *entity resolution* component resolves overlapping device and session records and groups events that likely originated from the same browser or client instance, with the qualification that the latter is an investigative aid, not definitive attribution (§ 4.2). Third, the *investigation user interface* displays normalized records through two views: the Events View shows a chronological table of records, and the Devices View shows resolved devices, sessions, and event groups (§ 4.3). Across both views, users can filter, compare, and annotate records with context.

We implemented LESTrADE as a static web application so it can be accessed from a personal or clinic computer,³ including during a remote consultation, without sending export contents to an application server [99]. After the initial design, we iteratively improved the interface using participant feedback (§ 6). Following Stephenson et al. [93], features added or substantially modified during the study are marked with  Update X and summarized in Appendix B. As a general principle, we made an update when at least two participants independently raised the same feedback.

4 Implementation

The user interface is implemented in Vue.js [105] using components adapted and substantially modified from the open-source Timesketch project [49]. The data import pipeline (ingestion → normalization → entity resolution) is implemented in Python and runs in the browser via Pyodide, a Python runtime compiled to WebAssembly [95]. LESTrADE temporarily stores files in the Origin Private File System (OPFS), a sandboxed storage endpoint in the local browser [3]. We use OPFS over IndexedDB [69], because it is better suited for large files and can run inside background Web Workers, which keeps the UI responsive during ingest. Figure 1 shows an overview.

³We currently deploy and host the static site via GitHub Pages, but it cannot be fully anonymized, so we do not link it here.

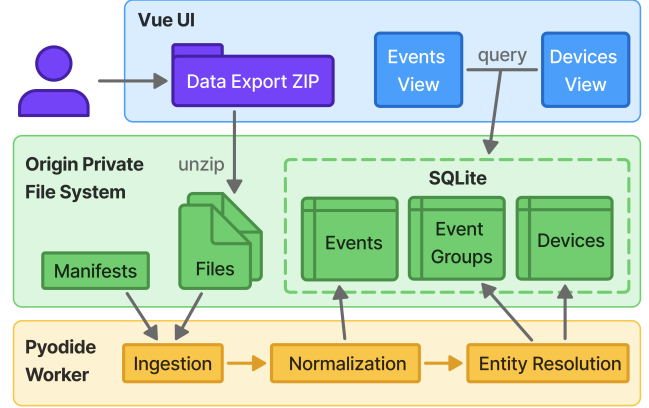


Figure 1: **Architecture of LESTrADE.** The application extracts selected files from a data export into the browser’s Origin Private File System. A Python/Pyodide worker ingests, normalizes, and links records in a local SQLite database, which is queried by the Vue interface.

4.1 Data Ingestion & Normalization

When a user imports a data export, LESTrADE stores it in the OPFS. The application extracts only text files listed in a platform manifest that we create; it does not extract media files. Keeping the archive, extracted files, and resulting database in OPFS allows the pipeline to run locally without blocking the interface. At the same time, LESTrADE starts a Pyodide instance in a background worker. The worker reads the selected files, parses their contents, normalizes the resulting records, and writes them to SQLite in OPFS using SQLite Wasm [97]. The Events and Devices views query this database rather than repeatedly parsing the original archive.

Declarative Manifests. One YAML manifest per supported platform specifies which files may be imported, how their keys or headers map to standard fields, and which semantic category applies to each record. Separating these mappings from the Python parser reduces the amount of code required when a platform changes a filename or field name. To support a new platform, it is usually sufficient to write a new manifest. The parser supports most CSV, JSON, and HTML and would only need changes for malformed files or unusual data formats. Field names follow Elastic Common Schema (ECS) conventions [38], e.g., raw fields “ipAddress” and “ip” both map to “client.ip”. The normalization pass also standardizes values like timestamps, user agents, and the names of operating systems and manufacturers. Each normalized record retains a reference to its original source file and line.

Semantic Categories. The manifests assign records to categories based on the taxonomy of “security-relevant” information identified by Nonnenkamp et al. [73]. They broadly distinguish between *events*, i.e., chronological actions, and records

describing the *state* of the account at the time the data export was prepared, such as a list of active sessions. Maintaining this distinction prevents a historical event from being presented as evidence that a session or device remains active. Categories of *event* records include "login" and "password_change," with optional outcomes such as "success" or "failure." Categories of *state* records pertaining to authenticated devices are described next.

4.2 Device Entity Resolution

Understanding which device generated an account event is useful during an investigation, but data exports rarely provide one stable identifier across all relevant records [73]. Instead, they contain two different forms of evidence. First, *state records* explicitly describe devices, sessions, or credentials known to the platform when the export was generated. Second, chronological *event records* may include only a user-agent string, IP address, or session identifier. LESTrADE handles these sources separately so that platform-reported entities are not conflated with relationships inferred by the tool.

Stable hardware identifiers would make linkage straightforward, but they are not consistently available and are rarely attached to every event. Hardware and operating system vendors such as Apple are more likely to report serial numbers or related identifiers, whereas other platforms often provide only a user-agent string. Apple also deliberately reduces identifying information in Safari user agents to limit fingerprinting [63]. Consequently, the system distinguishes deterministic linkage of state records from heuristic grouping of event records.

Deterministic Linkage of Device/State Records. *State* records related to authentication may describe several kinds of entities: recent, active, or trusted *sessions*; *registered hardware*, such as an Apple device associated with an iCloud account; *passkeys* with device or keychain metadata; unique *app installations*, such as an installation of Facebook or Messenger; or *platform fingerprints* used most often for advertising.

A single physical device may appear in several of these categories, but the categories are not interchangeable. For example, a passkey and an active session can both be associated with the same phone while representing different account relationships. To avoid false merges, LESTrADE links state records only when they share a deterministic *stable identifier* (session cookie, serial number, IMEI, or operating-system advertising identifier), which we call STABLEIDMATCH. When a registered hardware entry and a passkey share a serial number, for instance, the interface labels their association without erasing their original record types. Records without a shared stable identifier remain separate. This is performed within an individual data export, and LESTrADE does not attempt to identify the same physical device across platforms.

Heuristic Grouping of Event Records. Many event records lack a stable identifier. For these records, LESTrADE additionally constructs *event groups*: connected components of

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function EDGES( $R; \Delta t_{\max}$ ):
   $A \leftarrow \{\text{model, os\_name, os\_version, client\_name}\}$ 
   $E \leftarrow \emptyset$ 
  for all unordered pairs  $\{u, v\} \subseteq R$ :
    if STABLEIDMATCH( $u, v$ ):  $E \leftarrow E \cup \{(u, v)\}$ 
   $R \leftarrow \{r \in R : \forall a \in A \cup \{\text{client\_ver, ts}\}, r.a \neq \emptyset\}$ 
  for all groups  $G$  of  $R$  grouped by  $A$ :
    sort  $G$  by timestamp
    for  $i \leftarrow 1$  to  $|G| - 1$ :
       $u, v \leftarrow G[i], G[i + 1]$ 
      if RULES( $u, v; \Delta t_{\max}, A$ ):  $E \leftarrow E \cup \{(u, v)\}$ 
  return  $E$ 

function RULES( $u, v; \Delta t_{\max}, A$ ):
  return ( $\forall a \in A : u.a = v.a$ )
  and  $\neg$ HARDWAREIDCONFLICT( $u, v$ )
  and  $0 \leq v.\text{ts} - u.\text{ts} \leq \Delta t_{\max}$  // monotonicity check
  and  $v.\text{client\_ver} \geq u.\text{client\_ver}$ 

```

Figure 2: **Edge construction heuristic** for entity resolution of *event* records (§ 4.2)

events that appear consistent with the same browser or client instance. An event group is an investigative aid, not a claim that its records originate from a uniquely identified physical device or actor.

We model each event as a graph vertex and add edges using the procedure in Figure 2. Records that share a stable identifier are linked (STABLEIDMATCH, see above). When no stable identifier is available, we apply a heuristic (RULES) derived from rules 1 and 2 of the FP-STALKER browser-fingerprinting algorithm [101]. Rule 1 states that two records must share a constant set of attributes A : operating-system name and version, model or manufacturer,⁴ and browser or client name (e.g., Chrome, Facebook App). Rule 2 states that client versions increase monotonically (e.g., Chrome v1.0 $\rightarrow$ v2.0). Many fingerprinting attributes used by FP-STALKER, such as Canvas output and font lists, are not consistently present in data exports [73]. Our heuristic therefore uses only attributes that the exports commonly provide. To account for this, we add constraints to RULES that timestamps differ by no more than $\Delta t_{\max}$ days and hardware identifiers (e.g., serial, IMEI) do not conflict (HARDWAREIDCONFLICT).⁵

We do not apply the heuristic on every element pairwise. Instead, we group records by attribute set A , sort records within that group by timestamp, and add an edge only between adjacent records. After edge construction, union-find produces disjoint connected components, which we call “event groups.”

⁴Vastel et al. [101] includes a *platform* attribute in rule 1; we decompose this into *model* (including manufacturer, when available) and *client_name*. These are parsed from the user-agent string unless explicitly provided.

⁵The heuristic does *not* guarantee that an entire connected component is free of hardware ID conflicts; e.g., for three consecutive records (r_1, r_2, r_3) , if r_2 does not have any hardware ID, all three records could be connected even when r_1 and r_3 have conflicting hardware identifiers. We therefore flag components containing conflicts post-hoc and let users resolve in the UI.

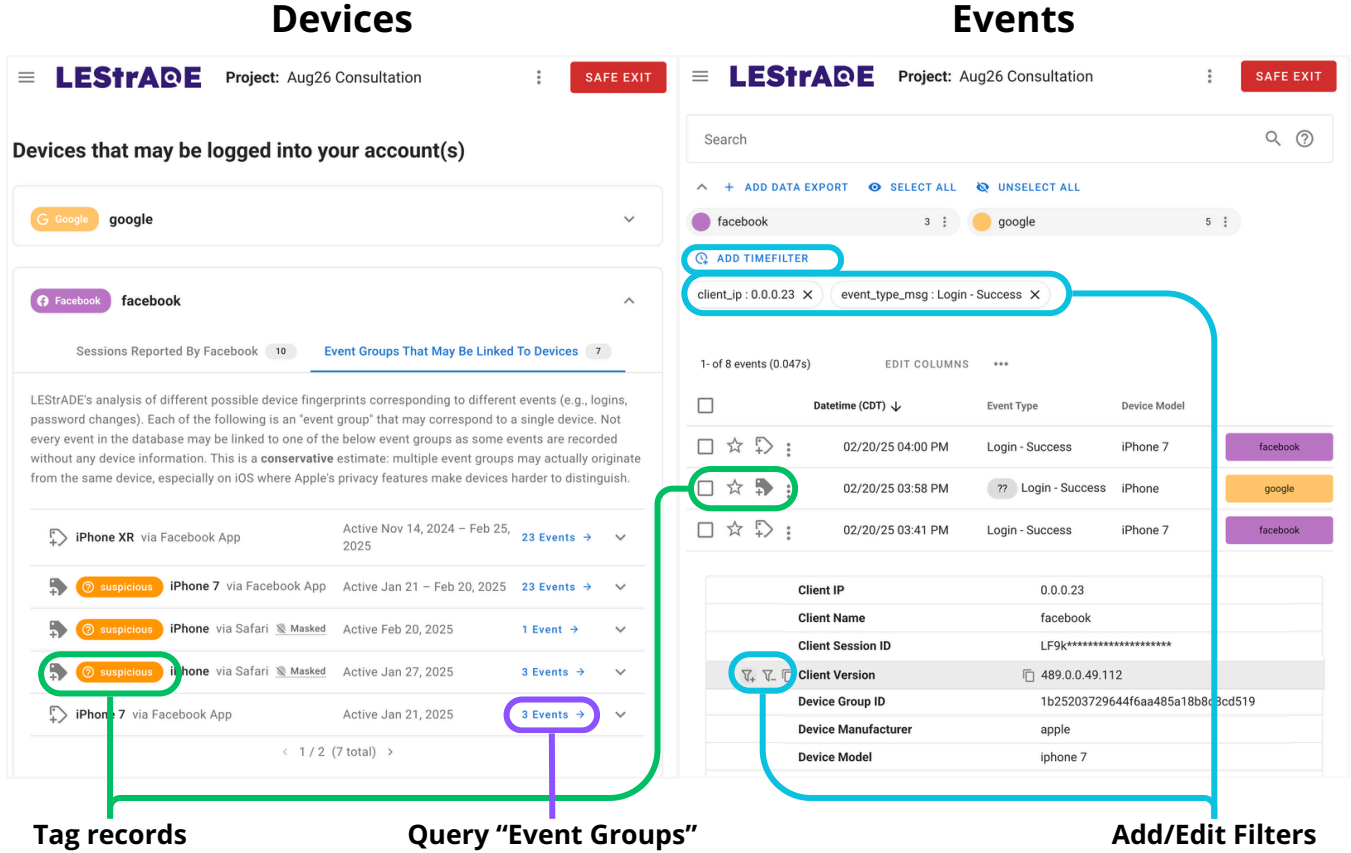


Figure 3: The **Devices View** (left) and **Events View** (right), along with a subset of key features: tagging both event and device records, querying groups of events from the Devices View, and multiple options for adding or editing filters by time or attributes.

Restricting application of RULES to temporally adjacent records makes the heuristic conservative. A spurious record with an older client version can split events that otherwise appear to belong to one client, reducing recall. Applying the rule to every pair, however, would not guarantee monotonic client versions throughout a resulting component and could introduce false merges. We prioritize precision because the two errors have different consequences in this context: a false merge may lead users to associate suspicious activity with a client they believe is safe, whereas a false split primarily creates additional groups that must be compared, which can be mitigated through the UI. In § 5.1, we evaluate this tradeoff and show experimental precision of 84–98% at $\Delta t_{\max} = 90$ days on real browser data without stable identifiers.

4.3 User Interface

The user interface of LESTRADE (Figure 3) is designed to guide user through the process of importing and analyzing the data exports. The landing page introduces LESTRADE, explains briefly what data exports are, provide guidance on how to obtain them, and an interactive tutorial on how to

use the tool. The page also prompts the user to import their data into the application. Users can import one ZIP file at a time by selecting a platform and adding the file either via folder selection or drag-and-drop Update 4. If the import is successful, users are first taken to the Events View, and from there, they can navigate to the Devices View (see Figure 3). Users can import additional data exports, which are incorporated additively into the existing views. At any point, users can visit a page with instructions for how to request data exports with the applicable categories or an interactive tour of important features with sample data.

Events View. The focal point of this view is a table where each row is an *event* record extracted from a data export, sorted chronologically (whose order can be toggled) with a default set of visible columns that the user can edit. Clicking on a row expands it to show all key-value pairs (“attributes”) associated with the record. The table components originate from Timesketch [49]. Users can add tags to each event row and pin up to five events to compare key-value pairs row-wise in a separate modal Update 6. To the left of the table is a collapsible panel that lets users filter by event type (“Login - Success” or “Failure”), IP address, and tags with dynamic

counts for each. Active filters become chips above the table that can be temporarily deactivated  Update 2 to let users switch between queries more easily. The top search bar also supports Lucene query syntax.

Devices View. This view attempts to help users identify the specific devices that generated logs within a data export. Each data export gets its own card with two tabs that map to different abstractions  Update 3. The first tab shows authenticated devices/sessions explicitly reported by the platform (see § 4.2, *device/state* records). The second tab shows “event groups,” or connected components with a shared set of attributes generated by the tool’s entity resolution heuristic § 4.2, *event* records).

Both tabs take the form of expandable table(s), similar to the Events View. If a row has an identifier that maps back to a set of events (e.g., session ID, event group ID), a button appears that takes the user to the Events View with the applicable query. Users can also add tags to device rows, with the option to propagate the tag to the corresponding events  Update 7. The interface briefly explains the limitations of “events groups” and attaches an information modal  Update 1 to records originating from Safari that contextualizes reduced user agents [63].

5 Computational Evaluation

In this section, we present findings on the performance of the data import pipeline: the accuracy of the entity resolution heuristic and the computational overhead.

5.1 Accuracy of Entity Resolution

We evaluate whether our entity resolution heuristic for *event* records (see § 4.2, Figure 2) reliably identifies groups of events that originate from the same browser or client.

Method We used the FP-Stalker dataset by Vastel et al. [101], which includes 15,000 browser fingerprint records collected from real users between 2015 and 2017 through the AmIUnique project [61]. Each original record contains a timestamp, user agent string, a long-term tracking cookie associated with the origin browser as ground truth, and several additional fingerprinting attributes (Canvas, font list) we discarded because they are not routinely available in data exports [73]. Each record’s user agent was parsed and normalized through LESTrADE’s pipeline to produce the device model, OS name and version, and browser name and version. We then use the entity resolution step (§ 4.2) to group these records and measure against the ground truth set of clients/browser inferred based on the long-term cookie. As the dataset contains no stable IDs beyond the long term cookie, all edges among the records are created using the RULES function (§ 4.2).

We assume that a single user (and hypothetical adversary) might have a small number (K) of devices which are used

to log into the user’s account. Therefore, we used sampling to approximate our anticipated use case. We consider $K \in \{2, 4, 6, 8, 10, 12\}$ devices or browser instances signed into their account. For each K , we drew K tracking cookies without replacement from two sets of cookies: (a) all unique cookies in the full two-year dataset and (b) unique cookies with at least one record in a $\pm N$ -day window ($N \in \{30, 60\}$) of a random date. We consider the (b) to test the situation where $\Delta t_{\max} > 30$ cannot trivially exclude inactive clients. we repeat 500 independent trials for each value of K . Then, we stripped the cookie/ground truth from the selected records, fed them through the entity resolution module of LESTrADE to produce connected components, and computed B^3 precision and recall over that set [12]. We repeated this experiment for different values of $\Delta t_{\max}$ between 7 and 180 days.

Possible limitations include that the random sampling procedure may not match the underlying distribution of the different device *types* that a typical user owns (e.g., one phone, one laptop). User agent reduction has also become more widespread since Apple first introduced it in Safari in 2017 [63, 108]. Thus, current exports may contain less information about device models or OS versions than the 2015-17 dataset. These limitations are more likely to impact absolute precision or recall than the general range of $\Delta t_{\max}$ that perform optimally (60–120 days).

Findings. In this context, B^3 precision measures that events from different clients/browser were not mistakenly merged into the same component. Recall measures that events from the same client/browser were not unnecessarily split into separate components [12]. F_1 scores over the parameter sweep are reported in Figure 4. Mean precision is greater than mean recall on every trial (see Appendix A). Precision decreases with increasing K (number of devices in the pool) and increasing $\Delta t_{\max}$. We choose a default parameter of $\Delta t_{\max} = 90$ days as it produced the highest mean F_1 (0.79) and $F_{0.5}$ (0.87; overweights precision) scores on the parameter sweep of the full dataset. At $\Delta t_{\max} = 90$, the mean precision on the full dataset, all K , was 0.94, ranging from 0.90 at $K = 12$ to 0.98 at $K = 2$. Under the windowed sampling regime with $N = 30$, precision at $K = 12$ fell to 0.84. Mean recall at $\Delta t_{\max} = 90$ was 0.68 on the full dataset, ranging from 0.63 at $K = 12$ to 0.77 at $K = 2$. Recall generally, but not strictly, rose with $\Delta t_{\max}$ as the parameter excluded fewer true matches, but declined with higher K , similarly to precision.

5.2 Performance

We evaluated the time and memory performance of the data import pipeline (data ingestion, semantic mapping/normalization, and entity resolution) in Pyodide by importing data exports of varying size into the user interface. We used Facebook and Google data exports generated by Nonnenkamp et al. [73], which contain 81 and 86 security records after ingestion, respectively. We created augmented data exports

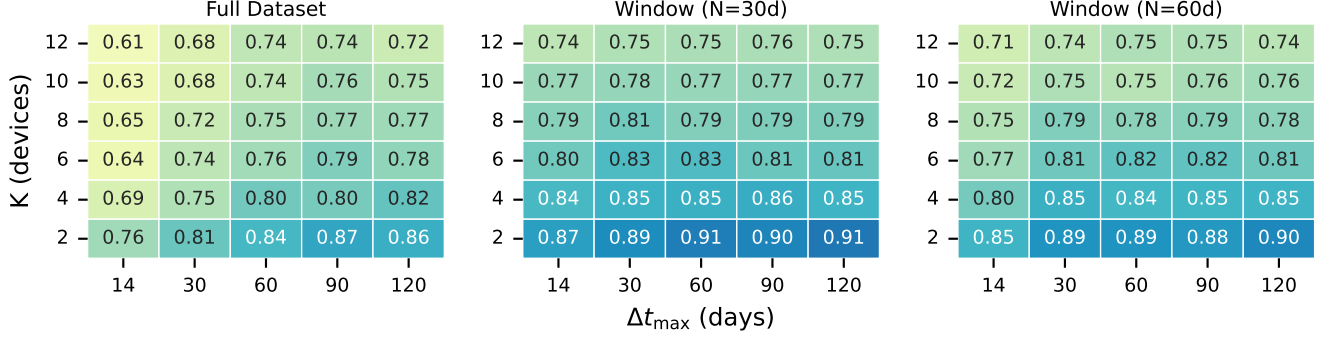


Figure 4: **B³ F1 score of Entity Resolution on subset of the AmUnique dataset dataset [61, 101] across sampling regimes.** The three charts below show the performance of entity resolution (§ 4.2) by sampling from (a) the full 2y dataset and (b) a random time window ($N = 30d$ and $N = 60d$). The heuristic performs better when differentiating between fewer devices (K). Low F1 is caused primarily by low recall, especially on lower, more conservative values of $\Delta t_{\max}$. Full sweep data ($\Delta t_{\max} = 7-180$) is given in Appendix A.

by replicating the records in each data export 10, 100, and 1000 times, with randomized timestamps.

Compressed data export sizes ranged from 0.13MB (original export) to 5.1MB (1000x augmented export). Contents are entirely text, since photos and media are redacted in the originals [73]. LEStRADE never unzips media files, only text files on the manifest whitelist. All trials were run using a local production build of LEStRADE via Google Chrome 151.0.7922.170 (arm64) on a 2022 MacBook Air M2 with 16GB RAM and a power supply. We ran three trials per data export (augmented and original) and polled the size of the Wasm linear memory used by Pyodide every 100ms at each pipeline stage.

Experimentation, albeit at a small scale, indicates that time and memory use of our pipeline scales feasibly with record size, as the 1000x augmentation represents an extreme case. LEStRADE can import the original data exports in 2.9s for Facebook and 3.8s for Google, with peak Wasm memory size of approximately 131MB for each. Pyodide’s baseline memory was 91MB of this. After multiplying the record count by 100, runtimes were 6.6s (2.3x increase over original) for Facebook and 20.2s for Google (5.4x), using 189.3MB and 188.6MB of memory, respectively (both 1.4x). At 1000x, entity resolution over Google took 63.1s longer than Facebook (172.6s vs 63.5s total). Likely, this was because Facebook provided explicit session IDs on most records, bypassing RULES (§ 4.2) while Google did not. Full evaluation data is shown in Appendix A.

6 Usability Evaluation

We evaluated LEStRADE in a user study with 11 technology clinic consultants, our intended user base. This study aimed to understand how consultants approach investigating an ac-

count compromise and how LEStRADE fits into that approach. Specifically, we sought to understand which strategies they used to resolve what kinds of uncertainty, usability friction and feature suggestions, and deployment considerations.

6.1 Method

We design a mixed-method, task-based user study [62] consisting of a structured task, a semi-structured interview, and a post-study survey. Two studies occurred in-person at the authors’ institution, and the remaining nine were conducted over Zoom. Participants visited LEStRADE in the browser on their own device (if Zoom) or on a Mac laptop provided by the researchers (if in-person). After obtaining consent, we used Zoom (regardless of format) to record audio and the browser tab showing the participant interacting with LEStRADE.

Study Procedure. We asked participants to consider a scenario in which they were leading a *mock tech clinic consultation* with a hypothetical survivor, Bob, whose ex-partner, Alice, had been logging into their Google and Facebook accounts to surveil their activity and send impersonating messages. We used data exports released by Nonnenkamp et al. [73], and our scenario and personas were designed to match their simulated account compromise protocol on research accounts. In the scenario, Bob had visited the tech clinic once before and had returned with their Facebook and Google data exports to learn more about the timeline and cause of account compromise.

The interviewer prompted users to freely explore the interface, then participants were given a document with necessary context about Bob’s and Alice’s history and devices (iPhone XR and iPhone 7, respectively). In this, they were given five investigation tasks to complete using LEStRADE and Bob’s data exports:

- Task 1:* Identify the time period Alice was logged in to Bob’s Facebook account.
- Task 2:* Identify any impersonating messages sent from Bob’s account or explain why unable to do so.
- Task 3:* Determine if the generic “iPhone(s)” logged in to Facebook, Bob’s, Alice’s, or a third device.
- Task 4:* Identify the time period Alice was logged in to Bob’s Google account.
- Task 5:* Identify any actions Alice possibly took while they were logged in to Google

Tasks served as a structured way for participants to familiarize themselves with LESTrADE. Participants often worked on multiple tasks simultaneously (i.e., 1 and 3; 4 and 5). If participants encountered difficulties completing tasks, the primary interviewer lightly probed about the features (or lack thereof) causing friction.

After tasks were completed, we conducted a semi-structured interview about features they found helpful or difficult, suggestions, and considerations for deploying the tool in a tech clinic setting. Finally, we administered a Post-Study System Usability Questionnaire (PSSUQ) [65] and a short demographics survey via Qualtrics. Our IRB reviewed the study protocol and materials and determined the study to be exempted from human subjects research requirements.

Recruitment. We contacted four U.S. tech clinics to request permission to post the study announcement via their email list or internal messaging channel. Participants had to be at least 18 years of age and have completed all training required to be considered a “consultant” at their tech clinic. We offered a \$25 Amazon gift card for their time. We recruited 11 consultants from three different Tech Clinics between March and July 2026.

Participant Characteristics Of 11 participants, six (55%) identified as men, three (27%) as women, and two (18%) as non-binary or gender-non-conforming. Five (45%) identified as white, three (27%) as Asian, and three (27%) as Hispanic or Latino/a. Most (73%) were aged 24-34 and all held at least a bachelor’s degree (73% had a graduate degree). Participants came from three different tech clinics in the U.S. Even though tech clinic model has existed for less than a decade, 27% of participants had at least five years of experience as a tech consultant and 55% had three or more. One had worked in IPV social services for more than 10 years.

Data Analysis We transcribed recordings and produced speaker diarization using a local instance of WhisperX [13], then cleaned and removed incidental identifying information. We analyzed the interview transcripts alongside the screen recordings using two separate coding approaches for two sets of codes (D1-3 and I1-4).

First, using deductive content analysis [86], we coded data for (D1) each feature used (LESTrADE does not collect click logs due to its privacy model), (D2) whether they used it naturally or after being prompted by the interviewer, and (D3)

Metric	PSSUQ Subscales			
	SysUse	InfoQual	IntQual	Overall
Mean M	2.00	2.16	1.70	1.98
SD	0.61	0.83	0.85	0.65
Norm. M^*	2.80	3.02	2.49	2.82
$t(df = 10)$	-4.35	-3.40	-3.10	-4.27
p	< 0.001	0.003	0.006	< 0.001
95% CI	[-1.21, -0.39]	[-1.42, -0.30]	[-1.36, -0.22]	[-1.28, -0.40]

Table 1: **PSSUQ subscale and overall scores (n=11).** Mean scores (M) compared to normative baselines (M^*) [65]. Lower scores indicate higher usability (scale 1–7). CI refers to $M - M^*$.

their immediate reaction to it (e.g., positive, negative, etc.). Two authors split coding, then double-coded two of eleven transcripts to compute Krippendorff’s α (with MASI distance for D1 due to multi-labeling) [11, 60]: $\alpha_{D1} = 0.874$, $\alpha_{D2} = 0.862$, $\alpha_{D3} = 0.658$. Because of low α_{D3} , we proceeded with frequency analysis only on D1 and D2. Second, we used structural coding with inductive subcoding [83] where the initial top-level codes matched the study goals: (I1) Investigation Practices, (I2) Uncertainty, (I3) Friction & Suggestions, and (I4) Deployment Considerations. One author developed an initial codebook using two transcripts, then a second author applied and extended the codebook on both. The authors met to clarify definitions, then repeated this on two more transcripts to resolve disagreements through discussion until consensus was reached [40].

Limitations. Because there are few tech clinics and consultants from which to recruit, sample size ($n = 11$) may limit the range of feedback. Our participants skew highly educated, which means our findings may understate difficulties (e.g., technical jargon) that a different user base might encounter. The “mock” aspect of the mock consultation may have softened how participants communicated real-world risks. This was a deliberate choice to gather feedback safely before piloting with a tech clinic, and we probed participants about how they might explain findings to a survivor in a real consultation. Participant task answers varied in precision and hedging in a way that is consistent with how uncertainty is communicated during tech clinic consultations [77]. The main focus of analysis is not binary task success; it is the strategies participants used to navigate data and their perspectives on deployment.

6.2 Usability, Friction, & Suggestions

PSSUQ. The Post-Study System Usability Questionnaire (PSSUQ), which participants completed at the end of the study, measures perceived usability from 1 (Strongly Agree) to 7

(Strongly Disagree) across three subscales **System Usefulness**, **Information Quality**, and **Interface Quality** [65]. We ran one-tailed, one-sample t-test ($df = 10$) against normative baselines (M^*) derived from studies averaging 10 participants each [65, 85], testing whether LESTrADE scored better (lower) than M^* . We show results by subscale in Table 1, and by question in Appendix B. Even though no participants had used LESTrADE before the study, they rated it as significantly more usable than the normative baselines on every subscale. We contrast this with the fact that all participants reported some degree of friction in the user interface and gave suggestions, detailed above. This suggests that the questionnaire captured participants’ satisfaction with what LESTrADE let them accomplish compared to a typical tech clinic investigation, and that friction felt addressable through design.

Friction within the UI. Participants encountered moments of friction with the user interface, many of which we resolved or patched between studies. In early versions, Event View filters stacked unintuitively: “*some people... might feel like, oh, so there’s no data, but no, it’s because there are two filters applied right now*” (P5) [Update 5]. Participants noted that the expandable event metadata might be overwhelming for less technical users and suggested making it “*more user-friendly by hierarchy*” (P8). Several reported difficulties remembering long strings and dates, with P7 explaining “*sorry, I lost track of which IP it was looking for, and this is why I’m clicking around vaguely.*” Several participants created ad-hoc ways to compare attributes across contexts by duplicating browser tabs in “*split view*” (P10). P6 alluded to “*you know how you go to a computer store or a phone store online, and it’s like, select two items to compare... and it just shows you which fields are different,*” which we later implemented [Update 6].

Constructive Suggestions. Participants also ideated about broader suggestions, for example, more ways to visualize relationships between events and devices. P5 asked, “*what if there could be a view, like a timeline, where you could easily see if there is an overlap between the usage or logins of certain devices?*” P10 wondered if they could “*mark a device as trusted*” and have that propagate to its associated events [Update 7]. Additional suggestions included “*annotat[ing] the timeline to have contextual information*” (P8), a “*glossary*” (P6) for terms, and a way to “*export*” (P9) the annotated data and “*provide it to the police or to the court*” (P11).

6.3 Investigation Practices with LESTrADE

Throughout the five tasks, participants relied on a number of investigative strategies. Three recurring high level strategies to locate activity associated with Alice’s (adversary) devices were: temporal reasoning, comparing device identifiers like model and OS, and correlating IP addresses between records.

Temporal Reasoning Participants used timestamps and

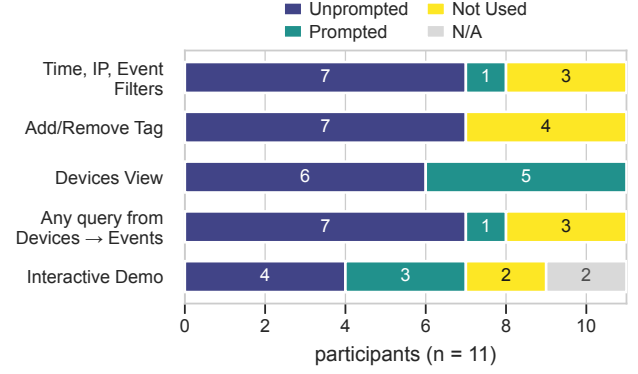


Figure 5: **Feature discovery and use (n=11).** Each bar shows how many participants used each feature(s) and, if so, whether that use was prompted by the interviewer or unprompted. N/A records mean that a feature was not yet deployed or working.

temporal patterns to establish when a device was active or attribute activity to an actor. For example, Task 1 (see § 6.1) asked when, if at all, the adversary’s device was signed in to the survivor’s Facebook account. Seven participants formed initial hypotheses by scanning or filtering the Events View table to find rows with the adversary’s device model (“iPhone 7”) to find bounding dates. The remaining four participants navigated to the Device View early on, where a card summarized the device’s activity: “*According to this, it was logged in from January 21st to February 20th*” (P8). Probed why they switched views, P5 reported that the Events View had “*too much information.... If I go to Devices, I automatically filter out the other information that I’m not interested in.*”

In Task 4, which asked participants to do the same for Google, P11 pointed to a cluster of consecutive login attempts in a short period as a sign that they were done by Alice, the adversary: “*A normal scammer doesn’t try that many times.... If you see a repeat [login], it’s because it’s someone specifically looking to get into your account only.*”

Comparison of Device Model and OS In Task 1, most authentication records in the data were associated with a precise device model (“iPhone 7” for the adversary) that participants could identify. This was not the case for Tasks 3 and 4, which asked participants to resolve ambiguity caused by user agent reduction in Safari (see § 4.2). In the ground truth data, two devices (iPhone XR and 7) accessed accounts through both apps and Safari [73], creating generic “iPhone” records that could have originated from either device. A few participants deduced the reason from prior knowledge: “*My thought would be if you log in via a browser, the user agent might just say, ‘this is an iPhone,’ not say ‘this is an iPhone 7’*” (P6). Others eventually moused over a help modal in the Devices View that provided context.

Without a model name to rely on, seven participants compared records using iOS versions present in both mobile app and Safari records: “*iPhone XR says [iOS] 17.7.1 and iPhone*

7 has 15.7” (P5), unless the survivor “downgraded the phone... or Alice figured out how to tamper with the user agent” (P10). P1 searched the internet to confirm that iPhone 7s are incompatible with iOS 17. After noticing the OS discrepancy, four participants revised earlier hypotheses, for example, P4 “assumed some of the ‘iPhones’ were probably still [the] iPhone 7, maybe even when they weren’t”.

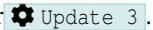
IP Address Correlation Eight participants checked if records with vague or no identifiers shared an IP address with less ambiguous records. “Facebook has more granular... device identification [than Google]. I’m just gonna try to store those IPs,” P7 announced, using a separate notepad to do so. This was useful for activity attribution during Task 5, as Google’s password reset records had no identifiers besides an IP: “Is it the same IP address as...? It’s probably most likely Alice that did the reset” (P8). Initially, P11 noted that since IP-derived “location is the same... as the previous [record], it’s pretty safe to say it’s probably the same person”, but then added the caveat that both users “would probably be in this close-ish area” due to the threat model. We expand on such caveats in the following section.

6.4 Uncertainty

Participants were encouraged to expand on the uncertainty about their *hypotheses* they encountered and (sometimes) resolved during the tasks. This tended to center around (1) the underlying data export and how platforms log user actions, (2) how LESTRADE has transformed or interpreted that data, and (3) the survivor’s external context that might not be present in the data at all.

Data Limitations Most participants noted the absence of static device identifiers like the “serial number” (P4) or “MAC address” (P1), as discussed in § 4.2. P8 expressed that “it would be better if Google and other service providers actually revealed more, if not all, of the information they have about the sessions, as in session cookies... or if they can’t, then there’s something wrong in their logging infrastructure.” This had implications for participants’ assessment of safety and risk, with P11 noting “it doesn’t show for the logouts who was logging out, so potentially the abusive partner might have not logged out.”

Participants sought clarity about which user or system actions each platform recorded and why. Several expressed that Facebook’s method of establishing sessions did not match “people’s working models of login” (P8). Looking at consecutive login and session refreshes from different devices, P7 pointed out “I also don’t know at this point if Facebook—when you log in from one place— affects the session in another.” P8 noted “I don’t know the format of these data logs, but it sounds like they’re ... ascribing the same session cookie to different logins, which is a little confusing to me, but maybe that’s how Facebook does it.”

LESTRADE’s Interpretation Uncertainty also emerged from how the tool itself parsed and displayed the data. Several participants asked the interviewer for clarity about the taxonomy of event types (§ 4.1) that LESTRADE had assigned to records in order to enable queries across multiple platforms: “The event types... are these the ones that are automatically generated based on the data in the export?” (P4). Participants were uncertain about how specific terms such as “authentication challenge” (P3, P4, P6) and “session refresh” (P6, P9, P10) mapped to concrete actions. Especially earlier in the study, participants seemed uncertain about how the Devices View grouped events by client (browser/app instance): “Does this group it all under one time?” (P3). After six participants, we pushed a significant update to the Devices View to make entity resolution visually clearer.  Update 3.

Several participants appreciated that the lack of automatic classification (§ 3.3) let them “make sense of [the data] as you will” (P1). “One of the things that I love... it makes no assumptions,” P6 added, “the scope of this is perfect.”

Survivor’s Context Participants often provided caveats regarding their initial hypotheses depending on the real-world lives of the hypothetical survivor and adversary. If they were living together, P8 noted that “under a normal home WiFi setup, they would actually have the same IP address, but that address could be rotating.” Moreover, “it could be [the survivor’s] device, but perpetrators can also— if they have physical access— they can use the survivor’s device as well” (P5). Participants also wondered aloud if “adversary has the technical capabilities to perform an IP spoof” (P1) or if they should be “concerned about VPNs” (P2). Ultimately, without this context, definitively attributing activity was difficult: “It’s always tough when it’s like, is this [the survivor’s password] reset or the adversary ... causing harm?” (P8).

6.5 Deployment Considerations

Benefits of LESTRADE. We asked participants about what value LESTRADE might bring to a tech clinic consultation. P7 appreciated the ability to ground survivors’ (or *clients*’) concerns in “something so concrete... when you don’t have any evidence that something is wrong and [you’re] trying to tell the client that in a trauma-informed way.” Determining the identity of the “mystery iPhone” (P6), a common frustration in prior consultations, was a task that participants were impressed LESTRADE could help them resolve. “I’ve never been able to [do it] that fast, [to] say this is an iPhone XR, this is an iPhone 7”, P1 mentioned. “The amount of times that a client has been concerned about the lack of information from the Google sign-in has been unbelievably traumatic for a lot of people.” Two participants hypothesized about the evidentiary viability of data exports: “If were to print out [LESTRADE] for a tech client of mine so that they could provide it to the police or to the court... that would be huge” (P11).

Potential Challenges. Participants anticipated some logistical challenges. First, requesting exports is time-consuming. P9 pointed out that “[the data] would need to be requested [in] that first conversation with the [survivor].... with the [consultant] just to make sure it’s as minimal as possible.” Extra consultations (see § 3.2) could impose a burden on tech clinics: “we do not have enough volunteers” (P9). P8 thought it might be “stressful” to navigate the tool while “trying to like communicate as you go with the client” during a time-limited session.

Interpretive Role of Consultants. Most participants agreed that, in its current state, LESTrADE would be most effectively used by a tech expert present to “interpret and communicate” (P2) results for a survivor. They suggested that to be more responsive to a general user base or “caseworkers or advocates that are [less] tech savvy” (P5), the tool would need more “information explaining what this data is” (P10) or a “beginner-friendly view.” (P2)

7 Discussion

Inconclusive Results & Traumatic Stress Under our operational protocol for tech clinics (§ 3.2), the survivors who would reach the stage of using LESTrADE are likely those who have already formed a suspicion about account compromise that the account security interface (ASI) has failed to either confirm or resolve. If (1) compromise occurred, (2) it is sufficiently logged in the data export, and (3) the consultant and survivor can identify it, then all is well. If not, it may not be possible to tell *which* of the three conditions has failed to hold. Under standard operating procedure for tech clinics, such results are generally considered inconclusive and not safe by default, so remediations (e.g., changing passwords) are undertaken regardless [43].

However, Ramjit et al. [77] note that inconclusive findings can trigger or reinforce traumatic stress reactions among survivors, including anxiety, grief, and hypervigilance [84]. A survivor might feel grief if they were hoping LESTrADE would give them evidence for an upcoming court hearing. For others, inconclusive results might conflict with “pre-conceived theories about tech abuse [that] may be protective” [77], and thus, they may keep requesting follow-up consultations or interpret ordinary records as threatening. This is a significant reason why we believe our operational protocol ought to keep a consultant in the loop beyond just interpreting technical jargon (§ 6.5). We see an opening for LESTrADE to provide additional language in the UI (similar to flagging Safari user agent reduction  Update 1) that could help consultants mediate these types of delicate interactions and demystify inconclusive results in a trauma-informed way.

Data Exports & Incidental Data Subjects. On the home page, we establish that LESTrADE is meant to do “consensual digital forensics,” a phrase we attribute to Amnesty In-

ternational’s Mobile Verification Toolkit [8], in which users willingly analyze *their own* data for their safety and security. However, as Walquist et al. [106] note, data exports contain the primary (presumably consenting) account holder’s data *and* a significant amount of data belonging to other people. Exports include conversations, friend requests, family sharing settings, and an amalgamation of other data that incidentally implicates others, whether or not they have consented. The fact that LESTrADE only processes security data by design mitigates privacy risks to the primary user and any incidental data subjects. The exception is when a third party appears within the security logs (e.g., an ex-partner adding their recovery email). However, these are typically critical records needed to protect the client’s safety and thus we include them despite the possible privacy tradeoffs.

Future Work & Development. Several features suggested by participants remain unimplemented, most notably a glossary for jargon, as we would like to test empirically what explanations are intuitive to non-technical users. Developing this is near-term future work. P9 and P11 raised the possibility of extending LESTrADE to legal contexts (i.e., evidence for survivors’ protective orders), similar to Sherlock [93], which could extend the tool’s impact but also introduces additional complexity about provenance and authenticity.

We purposely designed LESTrADE’s parser to be fairly easy to maintain. Updating fields or adding new platforms can mostly be done in YAML without touching the Python. We are exploring ways to let users optionally send us privacy-preserving diagnostics if the parser fails. Currently, we support only English-language files, as the public datasets available to us during development were limited to English and German [73, 76]. We observed that naming conventions in these differed even after translation; as such, building more robust mappings might require a data donation campaign to collect and design for multilingual exports.

8 Conclusion

We contribute LESTrADE, a browser-based visualization tool designed to let survivors of technology-facilitated abuse, as well as the technologists supporting them, investigate account compromise using account data exports. LESTrADE lets users filter, search, and annotate security-related records from multiple platforms at once. It does all data processing locally and applies an entity resolution heuristic that, in its default configuration, achieves 84–98% precision when grouping real browser records by their originating instance. In a user study with 11 technology clinic consultants, we find that LESTrADE is well-received and helps consultants make use of otherwise unusable data exports. We discuss practical and trauma-informed deployment considerations and future development plans.

Ethical Considerations

We consider ethical considerations both regarding the user study protocol (which was approved and determined exempt by our institution’s IRB; see § 6.1) and publication of our artifact.

Relationships & Power Dynamics in the User Study We considered the ethics of conducting research with the small community of technology clinics in the U.S. who support survivors of intimate partner violence. Many consultants within and between clinics have personal or professional relationships with one another and have communication channels where they share advice and resources. Several U.S. tech clinics are affiliated with academic computer science departments, which introduces power dynamics, many unspoken, about mentorship and hiring. Since tech clinic volunteers are LESTrADE’s intended initial user base, evaluation with this population was necessary. However, the research team had to take special care to make sure that participants (1) joined the study due to genuine interest or curiosity about LESTrADE, especially considering new consultants or graduate students with less relative power, and (2) felt free to critique the tool candidly. For (1), a junior graduate student obtained consent and was the primary interview facilitator to reduce the possibility of a professional power asymmetry between the research team and the participant. For (2), we explicitly invited constructive criticism of the tool at the start of each user study. Our protocol also included comparatively more questions about what could be improved than what already worked well Appendix B.

We know these steps cannot completely mitigate the above power dynamics. Still, participants provided extensive constructive criticism about UI features and challenges they might encounter during deployment (§ 6.2). We hope this is a reflection of the community’s interest in seeing tools like LESTrADE improve to fit their needs.

Potential Weaponization of LESTrADE We distinguish between the safety risks to survivors while obtaining data exports and the potential of the tool itself to be misused by an abusive partner. To address the former, we designed our operational protocol for technology clinics (§ 3.2) to mitigate this as best as possible by encouraging clients to request exports with an experienced consultant after guaranteeing their circumstances allow for safe request and storage of the sensitive files. For the latter, we consider an abuser who has breached a survivor’s account, requested the survivor’s data export, and now wants to use our tool to extract sensitive data more efficiently. Of the data LESTrADE exposes, only IP-derived location plausibly serves the abuser’s goals (§ 3.1). However, an abuser with account access likely already has easier means to obtain location information through ASIs or other account contents. We do not believe LESTrADE meaningfully adds to their capability.

Digital Forensics & Civil Liberties The authors’ current research context is the United States in 2026, where federal law and immigration enforcement increasingly uses social media and messaging data to surveil marginalized individuals, namely immigrants, and facilitate the suppression of lawful speech. U.S. agencies execute this either by purchasing or scraping bulk data or by submitting targeted search warrants to platforms for specific user records. Arguably, the latter is increasingly weaponized in order to chill free expression; for example, earlier this year, the U.S. Department of Justice tried (and failed) to obtain search warrants for YouTube account data belonging to journalists in Minnesota who were covering protests against Immigration and Customs Enforcement (ICE) [103]. It appears that many “warrant returns” (the file package returned from a search warrant) and data exports may not be so different. Recent journalism [100] and our preliminary analysis of the open-source warrant analysis tool RLEAPP [21] suggest that Google provides many of the same file types in both warrant returns and data exports. Thus there is a small— but non-negligible— chance that LESTrADE could be used for this purpose, most realistically by its data ingestion pipeline being subsumed into other warrant analysis tooling.

The research team objects to this (mis)use, but just objecting in the prose of a research paper is insufficient. We attempt to address this, imperfectly, through licensing. The parser and entity resolution code are released under a Creative Commons Attribution-NonCommercial-ShareAlike 4.0 license which prohibits use in commercial software or services. The user interface, which includes some modified components from Google’s Timesketch project [49], retains its original Apache 2.0 license as required.

Open Science

All source code, evaluation scripts, data exports, figure generation scripts, and supplementary material are available at an anonymized repository (<https://anonymous.4open.science/r/data-export-gui-10F4/>). The anonymous repository is frozen and includes a Dockerfile to build the web application locally and execute both computational evaluations (§ 5). We currently host a static site via GitHub Pages, but since it cannot be fully anonymized and is continuously updated, we do not attach it here. To protect participant privacy, coded transcripts are omitted. Similarly, individual PSSUQ scores and the spreadsheet in which we ran t-tests are omitted. The study protocol and final codebook are provided in `supplementary_materials`. Our semi-structured interview questions are also listed in Appendix B.

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A Computational Evaluation (§ 5)

This appendix reports full computational evaluation results.

Entity Resolution: Figure 6 and Table 2 correspond to the entity resolution evaluation in § 5.1.

Performance: Table 3, Table 4, and Table 5 correspond to the pipeline performance measurements in § 5.2.

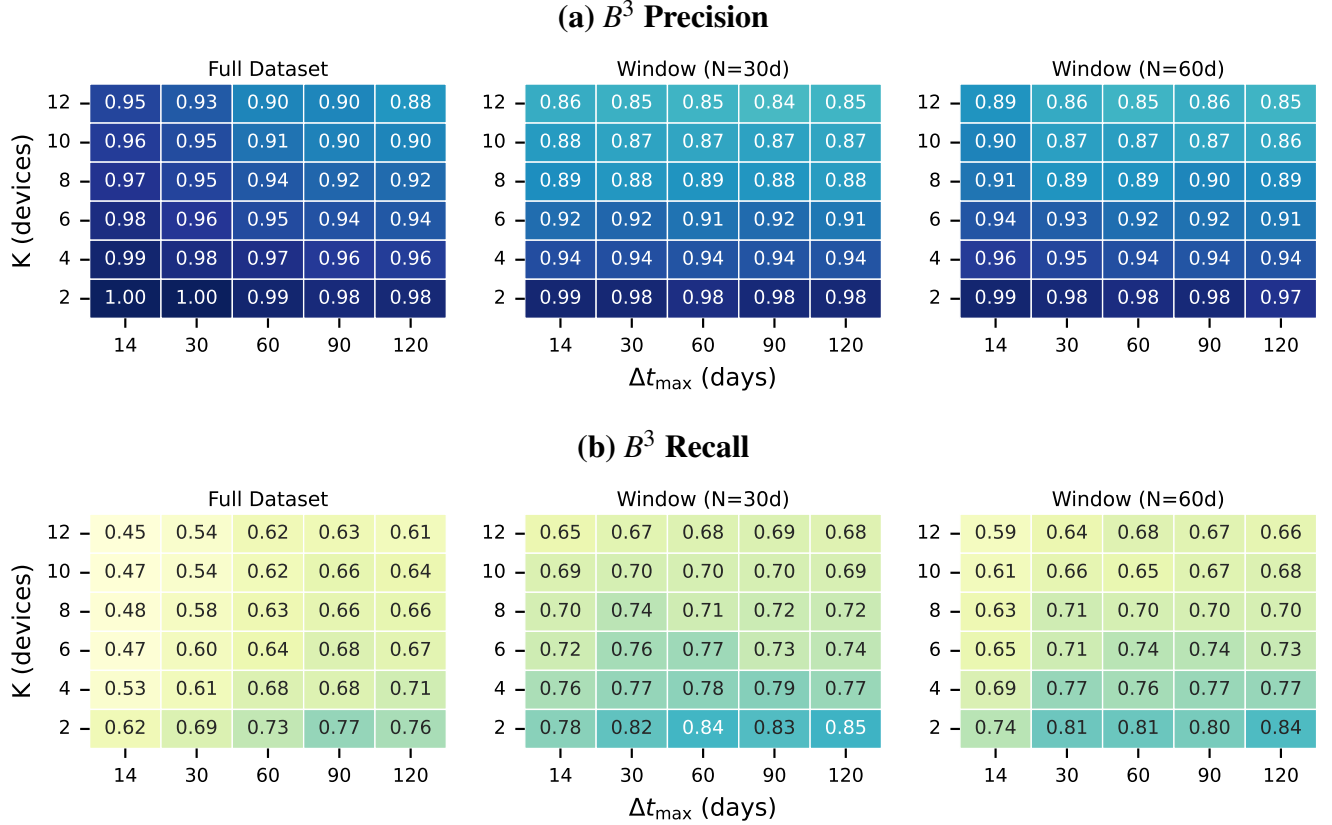


Figure 6: **Precision and recall sweep** across K devices and $\Delta t_{\max} \in \{14, 30, 60, 90, 120\}$ days, for all three sampling regimes. Full sweep range (7–180 days) is given in Table 2.

$\Delta t_{\max}$ (days)	Precision			Recall			F_1	$F_{0.5}$
	Mean	Mean Range	SD range	Mean	Mean Range	SD range		
7	0.982	[0.965, 0.998]	[0.019, 0.045]	0.425	[0.362, 0.544]	[0.154, 0.314]	0.591	0.775
14	0.973	[0.952, 0.996]	[0.035, 0.057]	0.504	[0.448, 0.620]	[0.184, 0.311]	0.663	0.819
30	0.962	[0.933, 0.996]	[0.033, 0.075]	0.592	[0.535, 0.685]	[0.212, 0.295]	0.732	0.854
60	0.943	[0.900, 0.987]	[0.061, 0.085]	0.654	[0.616, 0.733]	[0.219, 0.301]	0.772	0.866
90	0.935	[0.899, 0.981]	[0.081, 0.088]	0.681	[0.628, 0.775]	[0.210, 0.286]	0.787	0.870
120	0.930	[0.884, 0.983]	[0.080, 0.091]	0.673	[0.607, 0.756]	[0.219, 0.301]	0.781	0.864
180	0.922	[0.870, 0.983]	[0.080, 0.096]	0.680	[0.620, 0.761]	[0.218, 0.304]	0.783	0.861

Table 2: **Full parameter sweep.** Accuracy metrics on the full dataset, averaged across $K \in \{2, 4, 6, 8, 10, 12\}$ ($n = 500$ indep. trials per K). Mean range and SD range include the min/max mean and SD across each run of K . $F_{0.5}$ overweights precision per the design priority argued in § 4.2. F_1 and $F_{0.5}$ are optimal at $\Delta t_{\max} = 90$ days (bolded).

Platform	Multiplier	ZIP Size (MB)	Whitelisted Files, unzipped (MB)	All Files, unzipped (MB)
facebook	1x	0.13	0.06	0.27
	10x	0.06	0.61	0.77
	100x	0.17	6.09	6.26
	1000x	1.23	60.94	61.11
google	1x	4.00	0.19	11.01
	10x	3.98	0.61	11.40
	100x	4.09	4.76	15.55
	1000x	5.14	46.28	57.07

Table 3: **File sizes** of the original (1 $\times$) and augmented (10 $\times$, 100 $\times$, 1000 $\times$) data exports used to complete the evaluation in § 5.2. The “whitelisted” files are those specified by the YAML manifests as containing security information (§ 4.1) and the only ones extracted from the ZIP.

Platform	Multiplier	extract	semantic map	normalize	group (ent. res.)	total
facebook	1x	0.02 $\pm$ 0.00	0.02 $\pm$ 0.00	2.75 $\pm$ 0.04	0.13 $\pm$ 0.02	2.93 $\pm$ 0.06
	10x	0.07 $\pm$ 0.01	0.14 $\pm$ 0.03	2.82 $\pm$ 0.02	0.21 $\pm$ 0.03	3.27 $\pm$ 0.10
	100x	0.53 $\pm$ 0.01	1.20 $\pm$ 0.03	3.55 $\pm$ 0.03	1.29 $\pm$ 0.03	6.61 $\pm$ 0.03
	1000x	5.41 $\pm$ 0.08	12.99 $\pm$ 0.08	10.79 $\pm$ 0.10	34.26 $\pm$ 0.30	63.51 $\pm$ 0.05
google	1x	0.08 $\pm$ 0.00	0.03 $\pm$ 0.00	3.44 $\pm$ 0.05	0.19 $\pm$ 0.00	3.76 $\pm$ 0.04
	10x	0.49 $\pm$ 0.02	0.22 $\pm$ 0.02	3.52 $\pm$ 0.03	1.05 $\pm$ 0.01	5.30 $\pm$ 0.06
	100x	4.51 $\pm$ 0.07	1.88 $\pm$ 0.04	4.07 $\pm$ 0.03	9.66 $\pm$ 0.09	20.16 $\pm$ 0.21
	1000x	45.75 $\pm$ 0.34	19.81 $\pm$ 0.10	9.65 $\pm$ 0.03	97.38 $\pm$ 0.51	172.65 $\pm$ 0.45

Table 4: **Mean duration of execution (s)** ($\pm$ SD, $n = 3$) of each stage of the data import pipeline using both the original (1x) and augmented (10 to 1000x) data exports.

Platform	Multiplier	extract	semantic map	normalize	group (ent. res.)
facebook	1x	90.90 $\pm$ 0.49	90.90 $\pm$ 0.49	130.94 $\pm$ 0.69	130.94 $\pm$ 0.69
	10x	91.07 $\pm$ 0.31	91.07 $\pm$ 0.31	131.20 $\pm$ 0.41	131.20 $\pm$ 0.41
	100x	109.49 $\pm$ 0.38	131.40 $\pm$ 0.45	157.70 $\pm$ 0.53	189.29 $\pm$ 0.64
	1000x	271.76 $\pm$ 0.15	326.13 $\pm$ 0.19	326.13 $\pm$ 0.19	664.36 $\pm$ 0.30
google	1x	91.07 $\pm$ 0.31	91.07 $\pm$ 0.31	131.20 $\pm$ 0.41	131.20 $\pm$ 0.41
	10x	109.27 $\pm$ 0.38	109.27 $\pm$ 0.38	131.14 $\pm$ 0.45	131.14 $\pm$ 0.45
	100x	157.16 $\pm$ 0.11	157.16 $\pm$ 0.11	157.16 $\pm$ 0.11	188.61 $\pm$ 0.11
	1000x	809.89 $\pm$ 0.00	809.89 $\pm$ 0.00	809.89 $\pm$ 0.00	809.89 $\pm$ 0.00

Table 5: **Mean peak Wasm memory size (MB)** ($\pm$ SD, $n = 3$), by stage of the data import pipeline. The zero standard deviation for Google 1000x is because execution hit Pyodide’s maximum allocated ArrayBuffer ceiling. Since Wasm memory is linear, peak memory size over the whole pipeline is equal to that recorded after the **group (entity resolution)** stage.

B User Study (§ 6)

This appendix includes our interview questions, a table of the iterative design updates  Update X we made throughout the user study, and PSSUQ data and t-test results by question (results by subscale are reported in the main body, § 6.2). The full protocol is available on GitHub in supplementary materials. This also includes the PDF document we gave participants with background about Bob, Alice, and their devices for the hypothetical scenario.

Semi-structured Interview Questions

Tool-related questions - Usability & Friction

1. What do you like about this tool? What works well?
2. What do you not like about it? What should we change?
3. Do you have suggestions for how to change it?
4. Was any particular feature confusing to you? How can it be improved?
5. Was there any particular feature that stood out to you as helpful? What made it stand out?
6. Would you feel comfortable communicating the information synthesized by the tool to a client during a consultation? Is there any language we should improve?
7. Are there any features that might be helpful to add?

Task-related questions - Uncertainty

8. How relevant were the five tasks to an actual consult? Do you have any remaining questions about the tasks?
9. Were you able to complete the tasks? Did you face any challenges?
10. Was there anything in the UI that was confusing when you tried to complete any of the tasks?
11. Were there any features we could add that would have helped you complete any of the tasks?

Futuring questions - Deployment Considerations

12. Have you encountered situations in a consultation where this tool would have been helpful to resolve a client's questions?
13. Are there any other tasks you foresee this tool could help consultants with in the future?

Table 6: **Summary of UI updates** made during the user study in response to participant feedback. Updates were deployed between participant cohorts when at least two encountered similar friction or offered similar suggestions.

Participants	Updates made after these participants, but before the next.	
P1, P2	 Update 1	Safari Reduced User Agent Information Modal. Adds a hoverable chip next to Devices View records from Safari explaining user agent reduction [63].
P3, P4	 Update 2	Filter Chips. Converts active filters into interactive visual chips above the table. Chips can be temporarily disabled or permanently deleted, or turned into NOT(chip).
P5, P6	 Update 3	Major Devices View Redesign. Previously, hierarchy focused on model type (e.g., X records from iPhone 7, Y records from iPhone XR), but this abstraction was unclear and misleading. This explicitly demarcates what is parsed directly from exports the event groups inferred by LESTrADE in § 4.2.
P7,	 Update 4	Drag-and-drop File Import. Users can import ZIP files via selection in file system (previous) and drag-and-drop.
P8	 Update 5	
P9, P10	 Update 6	Smart Filter Stacking. If a new filter is added and there already exists an active filter of that same field type, the existing filter is temporarily deactivated to prevent stacking filter conflicts.
P11	 Update 7	Side-by-side Event Comparison. Users can pin up to 5 events from the table and launch a separate modal that compares attribute values row-wise and highlights differences.
		Device Tagging → Event Tag Propagation. Allows users to tag device rows and propagate them to linked events

Table 7: **PSSUQ Scores by Question (n=11)**. Mean scores (M) compared against normative baselines (M^*) [65] using one-tailed one-sample t -tests ($H_1 : M < M^*$) with variable df . Lower scores indicate higher usability (scale 1–7). CI refers to $M - M^*$.

Question	Mean M	SD	Norm. M^*	df	t	p	95% CI
Q1 Overall, I am satisfied with how easy it is to use this system.	2.55	1.44	2.85	10	-0.70	0.249	[-1.27, 0.66]
Q2 It was simple to use this system.	2.64	1.21	2.69	10	-0.15	0.443	[-0.86, 0.76]
Q3 I was able to complete the tasks and scenarios quickly using this system.	2.00	0.63	3.16	10	-6.08	<0.001	[-1.58, -0.74]
Q4 I felt comfortable using this system.	1.45	0.69	2.66	10	-5.81	<0.001	[-1.67, -0.74]
Q5 It was easy to learn to use this system.	2.18	0.98	2.27	10	-0.30	0.386	[-0.75, 0.57]
Q6 I believe I could become productive quickly using this system.	1.18	0.40	2.86	10	-13.76	<0.001	[-1.95, -1.41]
Q7 The system gave error messages that clearly told me how to fix problems.	4.00	1.79	3.70	5	0.41	0.651	[-1.58, 2.18]
Q8 Whenever I made a mistake using the system, I could recover easily and quickly.	2.63	1.51	3.21	7	-1.10	0.154	[-1.84, 0.67]
Q9 The information (such as online help, on-screen messages and other documentation) provided with this system was clear.	2.36	1.29	2.96	10	-1.54	0.078	[-1.46, 0.27]
Q10 It was easy to find the information I needed.	2.18	0.87	3.09	10	-3.45	0.003	[-1.50, -0.32]
Q11 The information was effective in helping me complete the tasks and scenarios.	1.27	0.65	2.74	10	-7.53	<0.001	[-1.90, -1.03]
Q12 The organization of information on the system screens was clear.	1.82	0.87	2.66	10	-3.20	0.005	[-1.43, -0.25]
Q13 The interface of this system was pleasant.	1.36	0.67	2.28	10	-4.51	<0.001	[-1.37, -0.46]
Q14 I liked using the interface of this system.	1.36	0.67	2.42	10	-5.20	<0.001	[-1.51, -0.60]
Q15 This system has all the functions and capabilities I expect it to have.	2.36	1.69	2.79	10	-0.84	0.211	[-1.56, 0.71]
Q16 Overall, I am satisfied with this system.	1.64	0.92	2.82	10	-4.25	<0.001	[-1.80, -0.56]